

Lesson Plan Template

Curvilinear Coordinates

Review and Bridge-In (1) Polar Coordinates (2) Jacobian (3) Change of variables

Learning Objective(s) By the end of this lesson, students will have demonstrated the ability to: 1. Compute Jacobian & line elements 2. Express div, curl, etc in various coordinates

Teaching Points and Content Input Strategies (Teaching)	Learning Activities (Application)
Consider the most efficient and active ways possible of delivering content. Introduce coordinate changes Cylindrical coordinates Spherical Coordinates ↕	Use strategies that get learners interacting with or applying content and skills. $(x, y) \leftrightarrow (r, \theta)$ $(\hat{i}, \hat{j}) \leftrightarrow (\hat{e}_r, \hat{e}_\theta)$ $\frac{\partial(x, y)}{\partial(r, \theta)}, \vec{\nabla}, \vec{\nabla} \cdot \vec{v}, \vec{\nabla} \times \vec{v}, \nabla^2 f$ $\iint dx dy \leftrightarrow \iint r dr d\theta$ $ds^2 = dr^2 + r^2 d\theta^2$ ↕

Review and Formative Assessment Ass 9b problem 18 Ass 10 problem 2

Closure and Link to Outcomes and Summative Evaluation Ass 10 Problem 1

